

Robust Visitor Counting by Duplicate Face Detection

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Abstract

We present a novel duplicate face detection method inspired by a psychological memory model that explains the human cognitive process through sensory memory, short-term memory, and long-term memory to improve the accuracy of visitor counting. To achieve this, we propose a detection array (DA), a short-stay queue (SQ), and a long-stay queue (LQ), corresponding to the three memory types in the cognitive model. Two deep learning models, MobileNet and ResNet, are utilized to detect face areas and extract features, respectively. Detected faces in the current video frame are associated with the registered faces in the DA to identify counting targets. Subsequently, duplicate face removal was performed by sequentially matching the target faces with the registered faces in the SQ (managed by stay time) and the LQ (managed by queue length). Unmatched target faces are newly registered in the SQ, and the visitor count was then updated accordingly. In the experiments, the proposed method was evaluated using the QMUL and ChokePoint datasets. The experimental results demonstrate improved counting performance, with the proposed method achieving 100% accuracy for the QMUL dataset and 94.82% accuracy for the ChokePoint dataset.

1. Introduction

Visitor counting systems aim to estimate the number of individuals entering and exiting a specific area. Accurate visitor counting is essential for developing service strategies that enhance customer satisfaction, and it has applications in various domains, including customer relationship management [1], retail management [2], urban planning [3], and video surveillance [4]. Traditional infrared signal analysis-based counting systems offer the advantages of low cost and fast response times [5]. However, they struggle to collect accurate data when multiple people pass through the infrared region simultaneously. Therefore, detection and tracking algorithms are particularly effective in addressing duplicate counting issues in computer vision, as they can continuously track individual attributes.

In vision-based visitor counting, two frameworks, the region of interest (ROI) and the line of interest (LOI), are

commonly employed to improve accuracy [6, 7]. ROI-based systems [6] estimate the number of visitors within a specific region, whereas LOI-based systems [7] focus on counting individuals who cross a predefined line in a video. Moreover, recent advances in face recognition (FR) techniques have enhanced the reliability of counting systems and expanded their applications [8]. Detected faces in consecutive frames can be identified using facial features extracted from FR models, reducing duplicate counts and ID-switching errors. To assign the correct face ID, two types of features are utilized [9, 10]: handcrafted features and deep learning-based features.

Also, visitor counting systems require an efficient data processing strategy to balance the trade-off between retaining old data and adopting new data. To address this issue, several long-term feature storage methods [11] that utilize cognitive psychology models [12] have been explored with promising results. The human cognitive process involves neural activity that allows the brain to receive sensory inputs, store knowledge, and solve problems [12]. For example, humans can recognize the same object over time by preserving visual features such as shape, color, and texture. The interaction between long-term storage and short-term storage enhances problem-solving accuracy, but its use in visitor counting remains challenging due to difficulties in maintaining data consistency issue when registering new feature vectors.

In this paper, we propose a novel multi-queue management-based duplicate face removal strategy inspired by the human cognitive process to enhance visitor counting accuracy. To achieve this, a sequential information processing model is designed based on psychological memory structures for efficient data storage and management. That is, the detection array (DA) stores detected faces from the input video, the short-stay queue (SQ) is utilized for the initial registration of counting target faces, and the long-stay queue (LQ) functions as long-term storage. Counting is performed through multiple face detection, followed by association and matching to prevent duplications. A newly detected face, once selected as a counting target through continuity analysis, is compared with the registered faces in SQ and LQ. The queue management strategy ensures consistent updates of visitor data. Experimental results demonstrate that the proposed

multi-queue management strategy significantly improves visitor counting performance in CCTV environments.

2. Related Works

Vision-based counting systems rely on algorithms to detect and track individuals by analyzing video captured in CCTV environments. Chen *et al.* [7] proposed a method to enhance face detection performance through edge pixel extraction and color space normalization. They estimated the number of people by tracking their center coordinates. These counting systems, which mainly depend on facial positions, struggle to achieve high accuracy due to the lack of biometric feature analysis. To ensure robust visitor counting, some systems have integrated end-to-end learning-based FR models with detection and tracking algorithms [8, 10]. Song *et al.* [8] employed label smoothing to prevent overfitting when training FR models using facial images collected from cameras in multiple stores. Autade *et al.* [10] developed a student counting system that compares each feature vector with the average feature vector obtained from analyzed enrollment face sets. Thus, the utilization of detection, tracking, and recognition techniques is important for robust visitor counting systems.

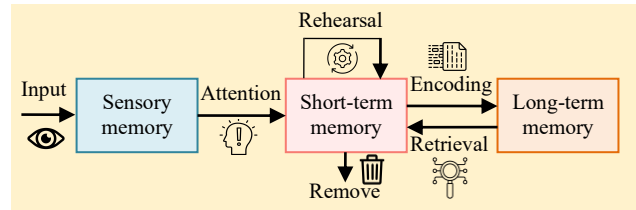


Figure 1. Diagram of the Atkinson-Shiffrin model.

Moreover, humans have a neural mechanism that allows them to distinguish objects by memorizing useful visual cues [12]. Atkinson and Shiffrin [12] proposed a multi-store model that describes human memory as comprising sensory memory, short-term memory, and long-term memory. The information processing flow in the Atkinson-Shiffrin model is illustrated in Figure 1. Sensory memory temporarily stores information acquired from external stimuli. Only attended information that receives attention is transferred to short-term memory for further cognitive processing. For long-term storage, frequently encountered information is encoded and then permanently stored in long-term memory. Long-term retrieval enables comprehensive analysis by combining stored knowledge in long-term memory with activated information from short-term memory. In visual object tracking approaches based on the Atkinson-Shiffrin memory model, Hong *et al.* [11] proposed a multi-store structure for processing short-term and long-term data. However, this tracker is optimized to identify predefined objects in the first frame. Therefore, its use in visitor data management is challenging in dynamic scenarios where new individuals continuously

enter a specific area.

3. Methodology

Visitor counting requires flexibility to enroll new inputs while preventing duplicates. To improve its robustness, this study utilizes a state diagram derived from the information processing mechanism of the Atkinson-Shiffrin model [12]. The proposed state diagram for handling facial information is illustrated in Figure 2. Three components, which correspond to the multi-store model, are defined: detection array (DA), short-stay queue (SQ), and long-stay queue (LQ). These components are designed to store, register, and manage facial information. First, DA temporarily registers detected faces in the input frames before considering them as counting candidates. Faces stored in DA are converted to a countable state when they receive sufficient attention through continuity analysis. Next, the duplicate status of each counting target is determined by sequentially searching SQ and LQ.

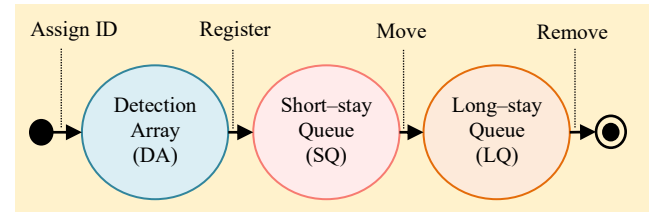


Figure 2. The state diagram for processing visitor face information based on the multi-store model. Detected face information in the input video is stored in one of the three storage components: DA, SQ, or LQ. The visitor count increases only when a face is initially registered in SQ.

After searching, the target face that remained countable is classified as a new visitor and subsequently registered in SQ, which is managed based on a limited stay time. Faces registered in SQ are transferred to LQ when they exceed the stay time, and LQ prevents overflow by sequentially removing facial information whenever the predefined queue length is exceeded. Additionally, LQ can pre-register resident individuals to exclude them from the counting results and store long-term visitor data. This structured approach is expected to improve counting accuracy through the separate management of frequently appearing faces or faces that are unlikely to reappear. Moreover, the proposed method reduces the complexity of face matching in the integrated feature space by optimizing the management of search targets and processing time to enhance duplicate detection efficiency. The overall visitor counting process is illustrated in Figure 3 and consists of three stages: *i*) multiple face detection, *ii*) face association for selecting counting candidates, and *iii*) face matching and counting using the multi-queue management strategy. The detailed techniques used in the visitor counting procedure are discussed.

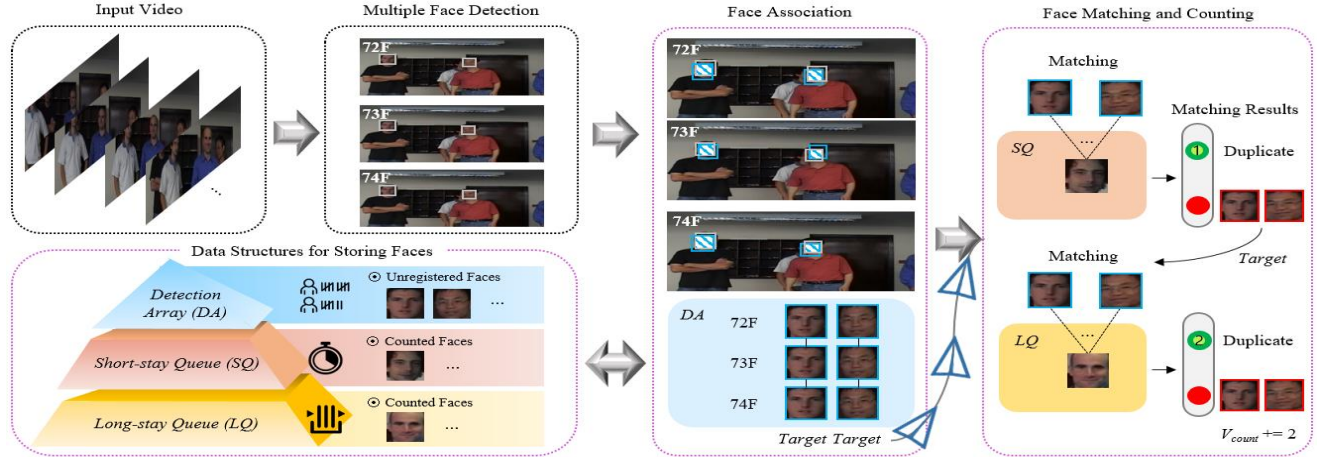


Figure 3. Multi-store model-based visitor counting process: Detected faces are associated and selected as counting candidates, while those similar to already registered faces are removed. To efficiently manage the accumulated registered data, SQ and LQ are maintained based on their stay time and queue length, respectively.

3.1. Multiple Face Detection

Detecting multiple faces from image sequences is the first step in counting visitors. In this study, we utilize RetinaFace [13] due to its high confidence scores and fast performance in detecting facial regions under challenging conditions such as varying scales, poses, and occlusions. After selecting target faces, each face image was normalized to the same size for feature extraction. We employed AdaFace [14] as the feature extractor, which demonstrated promising face recognition accuracy using an adaptive loss function. To reduce computational overhead in visitor counting, we adopted MobileNet0.25 [15] and ResNet18 [16] were adopted as backbone networks for face detection and feature extraction, respectively.

3.2. Duplicated Face Removal Strategy

A common visitor counting method manages all information in a single data structure with a large memory space. However, as more visitors are registered, the accumulation of similar features leads to a decline in new visitor counting. A multi-queue data structure-based method addresses this issue by processing data sequentially in the order in which it is received. That is, it stores facial data across multiple queues and manages them through interconnection. Consequently, this method is expected to improve matching and counting accuracy by enabling quick comparisons among registered faces and efficiently removing older data. The proposed duplicate face removal-based visitor counting method is presented in Algorithm 1. The algorithm consists of two main stages.

Phase 1 (Detected Face Association): For each frame f_t of the input video V , all faces D_t are detected at time t . Detected faces are stored in DA as U_t , where each face is assigned a unique ID and an *attention* value (η) for continuous analysis. The association is performed only

when registered faces existed in DA . A Kalman filter [17] was employed to estimate the positions of previously detected faces in the current frame, and the intersection over union (IoU) score [18] is utilized to measure the overlap between detected faces D_t and predicted face areas. The face association result O_t is used to increase the attention of DA , and newly detected faces are added to DA . Faces in DA that exhibit attention exceeding a predefined threshold are selected as counting candidates C_t and sent to *Phase 2*.

Phase 2 (Queue Management and Counting): This phase focuses on accurately counting visitors by analyzing the duplication status of C_t . The efficiency of duplicate detection can be enhanced by registering and matching C_t with stored faces in SQ and LQ , along with interactive queue management. First, the cosine similarity is calculated to determine whether C_t is already registered in SQ . Simultaneously, each registered face in SQ is matched only with the detected face that exhibits the highest similarity. Next, the same search approach is applied to LQ when C_t is absent from SQ . After confirming that C_t has low similarity to faces in SQ and LQ , the target face is registered in SQ , and the count is incremented. Finally, facial data that exceeds the stay time τ is moved from SQ to LQ , and old faces are removed sequentially to maintain stable storage space when LQ surpasses the queue length ω .

The similarity threshold for determining whether a detected face is newly appearing or already registered is established based on the Equal Error Rate (EER) analysis [19], which is commonly utilized in biometric systems. This threshold is calculated by considering the diversity of CCTV environments, such as lighting conditions, camera angles, and resolutions. Two queue management values, τ and ω , can be adjusted based on the number of visitors passing through the monitored area per hour. The purpose of this method is to improve duplication removal accuracy by sequentially processing visitor data.

Algorithm 1: Visitor Counting with Removal Duplicated Faces

Initialization: $DA \leftarrow \emptyset, SQ \leftarrow \emptyset, LQ \leftarrow \emptyset, V_{count} = 0$ **Input:**Each frame of video sequence V : $V \leftarrow [f_1, f_2, \dots, f_n], t = \text{index of frame}$ **while** V has not ended, **do**:

Detect face regions in current frame:

 $D_t \leftarrow [d_1, d_2, \dots, d_n], i = \text{index of detected face region}$

Phase 1 – Detected Face Association

Assign new ID to each unregistered region of D_t $U_i \leftarrow [u_i = (d_i, ID(d_i), \eta(d_i)) \mid \forall d_i \in D_t], \eta = \text{attention}$ $DA \leftarrow [u_1, u_2, \dots, u_j], j = \text{index of unregistered face region}$ **if** $DA = \emptyset$: $DA \leftarrow U_i$ **else**:Analyze face association result O_t to increase η of DA **if** D_t is not empty: $DA \leftarrow DA \cup U_i$ Select candidate face set C_t by checking η for each element of DA $C_t \leftarrow [u'_1, u'_2, \dots, u'_k], k = \text{index of target face regions}$ Send the face set C_t to Phase 2

Phase 2 – Queue Management and Counting

 $SQ \leftarrow [s_1, \dots, s_p], p = \text{index of registered face}, \tau = \text{stay time}$ $LQ \leftarrow [l_1, \dots, l_q], q = \text{index of registered face}, \omega = \text{queue length}$ Matching between each element of C_t and SQ **if** C_t not exist in SQ :Matching between each element of C_t and LQ **if** C_t not exist in LQ :Register target face C_t into SQ and increment V_{count} **if** $\tau > T_\tau$: // SQ management strategyMove registered face from SQ to LQ **if** $\omega > T_\omega$: // LQ management strategyRemove registered face from LQ **end while**

4. Experimental Results

The experimental setup included of an Intel Xeon processor, 1,024GB of RAM, and an NVIDIA RTX 6000, along with the PyTorch deep learning framework and the OpenCV library. To evaluate the performance of the proposed method, two public CCTV datasets, QMUL [20] and ChokePoint [21], were used; these datasets contain scenarios where people enter or exit within a defined camera field of view (FoV). However, these datasets lack sufficient ground-truth labels to analyze visitor counting performance. Therefore, experiments were conducted using images in which unique IDs were manually assigned to detected faces. Moreover, performance improvements were measured in terms of identification F1 (IDF1) and the number of predicted faces [22].

4.1. Datasets

Experiments were conducted using two RGB datasets that reflect changes in visit duration, movement paths, and speed at specific locations. The QMUL dataset [20] was recorded from a fixed camera viewpoint with a resolution of 720×576 at 25 frames per second (FPS) and consists of

three types of videos: Fast (S), Frontal (F), and Turning (T). The Fast category exhibits rapid visitor movements, the Frontal category captures front-facing postures, and the Turning category includes rotational behaviors. This dataset features up to four individuals, and has four variations: illumination, partial occlusions, head poses, and disappearance. A disappearance occurs when a person leaves the field of view (FoV) or is completely occluded by other objects.

The ChokePoint dataset [21] provides videos recorded at 30 FPS, capturing subjects as they enter (E) through Portal 1 (P1) and Portal 2 (P2) from three different camera viewpoints (C). First, P1E monitors the random movement of 25 individuals in a narrow corridor, offering four scenarios based on camera perspectives. Next, P2E includes scenarios where many people visit a wide area simultaneously, with five scenarios for each camera angle. In the P2E dataset, a total of 29 individuals participated, and brightness variations over time were observed. Compared to the QMUL dataset, the ChokePoint dataset has a higher resolution of 800×600 and features a larger number of subjects. A few original samples used in visitor counting experiments are shown in Figure 4.



Figure 4. Sample image for visitor counting and IDF1 analysis.

In this study, visitor counting was performed under the assumption of duplicate face detection, focusing on datasets captured from frontal viewpoints to extract more informative facial features. In the QMUL dataset, all three types of videos were used for performance evaluation. However, in the ChokePoint dataset, the P1EC3 video set, which consists of non-frontal images, was excluded from the experimental analysis. In addition, two previously unannotated subjects in the P1EC1 and P1EC2 video sets were included for counting purposes. This adjustment ensures that individuals moving through multiple routes are considered actual visitors. A summary of the datasets used in the experiments is shown in Table 1.

Table 1. Summary of datasets used in experiments.

Dataset	Type	# of Frames	# of Det. Faces	# of Max. Visitors
QMUL [20]	Fast (S)	487	1,186	3
	Frontal (F)	1,276	4,250	4
	Turning (T)	1,006	3,326	4
Choke-Point [21]	P1EC1	9,487	6,632	27
	P1EC2	9,487	5,350	27
	P2EC1	16,407	12,746	29
	P2EC2	16,407	15,518	29
	P2EC3	16,407	11,761	29

4.2. Experiment and Analysis

Detected faces were resized to 112×112 pixels, and 512-dimensional feature vectors were extracted. To determine duplicate faces based on a similarity threshold, offline EER analysis was conducted on the QMUL and ChokePoint datasets. In the experiments, the average IDF1 for each video was measured following the metric used by Pernici *et al.* [23], and counting accuracy was determined by calculating the average number of visitors estimated in each video. The comparative results of the proposed method are shown in Table 2.

Table 2. Comparative results of visitor counting and IDF1.

Datasets	Vid.	Algorithm	τ (s)	IDF1 (%)	Count. (%)
QMUL [20]	SFT	Pernici <i>et al.</i> [23]	–	93.40	–
		Zhang <i>et al.</i> [24] with FR	–	97.12	100
		Proposed method [†]	10	97.12	100
			12	97.12	100
Choke-Point [21]	P1E	Zhang <i>et al.</i> [24] with FR	–	56.36	86.60
		Proposed method [‡]	10	62.76	94.82
	P2E		12	60.93	92.91
		Zhang <i>et al.</i> [24] with FR	–	30.55	60.61
		Proposed method [‡]	10	31.59	63.37
			12	31.01	62.22

[†]: Length ω was set to 2, [‡]: Length ω was set to 8

In the first experiment using the three types of videos from the QMUL dataset, the IDF1 of the proposed method was compared with the results of identity learning approach introduced by Pernici *et al.* [23], which assigns a unique ID by measuring feature similarity based on reverse nearest neighbor estimation. To ensure a fair comparison with the object association method proposed by Zhang *et al.* [24], the experiment was performed by integrating the FR model that uses a simple storage space without enforcing queue management constraints. The proposed method achieved the highest IDF1 of 97.12% and counting accuracy of 100%. Compared to the results of Pernici *et al.* [23], the IDF1 improved by 3.72%, while achieving the same results as the method of Zhang *et al.* [24]. This result indicates that the

interaction effect of the proposed multi-queue storage structure is limited, due to the integration of the FR model and the small number of people appearing in the QMUL dataset.

The ChokePoint dataset contains more than 20 visitors, and experiments were performed on two video sets (P1E and P2E). In the first experiment, using the eight videos from the P1E set, the method of Zhang *et al.* [24] achieved 56.36% identification accuracy and 86.60% visitor counting accuracy. The proposed method defined the LQ length as 8, and the stay time (τ) for SQ was set to either 10 or 12 seconds for comparative analysis, considering the number of individuals appearing in the dataset. The experimental results showed slight performance differences based on the stay time of SQ with IDF1 improving by 1.83% and visitor counting accuracy increasing by 1.91% when the stay time was set to 10 seconds. Consequently, the proposed method achieved up to 6.40% higher IDF1 and 8.22% better visitor counting accuracy than the method of Zhang *et al.* [24]. To ensure that the proposed method also improves performance in crowded scenarios, similar experiments were conducted using the P2E set, which contains 15 videos with the highest number of visitors in the ChokePoint dataset. Although the performance on the P2E set was lower than on the P1E set, the highest IDF1 and visitor counting accuracy were achieved when the stay time of SQ was set to 10 seconds, consistent with the P1E results.

Experimental results show that adjusting the stay time of SQ and the queue length of LQ according to the visitor counting environments can significantly enhance performance. As such, the proposed method successfully minimizes counting and identification errors that commonly occur due to extensive comparisons of similar faces within a simple storage space. In other words, the structured management of facial information across multiple storages offers advantages in reducing duplicate detection complexity by efficiently removing or distributing comparison targets. Therefore, the proposed method is expected to improve counting accuracy not only for facial data but also for other types of objects. Moreover, storage space utilization can be enhanced through the dynamic management of new data and the automatic removal of less important data in a resource-limited memory structure.

5. Conclusions

We have described a multi-queue management approach inspired by a cognitive psychology memory model to improve face detection-based visitor counting accuracy in CCTV environments. To achieve this, a multi-queue storage management method was proposed, dividing visitor facial information processing into a detection array (DA) for attention tracking, a short-stay queue (SQ) managed by stay time, and a long-stay queue (LQ) managed by queue

length. In addition, a duplicate face detection strategy was implemented through interactive management between multi-queue storage and the FR module. Experimental results demonstrate that the proposed method improves the IDF1 by up to 6.40% and visitor counting accuracy by up to 8.22% compared with previous methods that utilize a simple facial feature storage space. Future studies on scheduling algorithms that can optimize multi-queue management by adapting to variations in visitor density are needed to solve ID-switching and undercounting problems in crowded scenarios. Furthermore, functional improvements are necessary to extend visitor counting applications toward customer relationship management and store management through additional analysis of detected faces, such as age, gender, and expression recognition.

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